The AI & Machine Learning Tree

Math | Classical ML | Deep Learning | Applied AI | MLOps | Responsible AI



6	28	121	39
branches	fields	tools	links

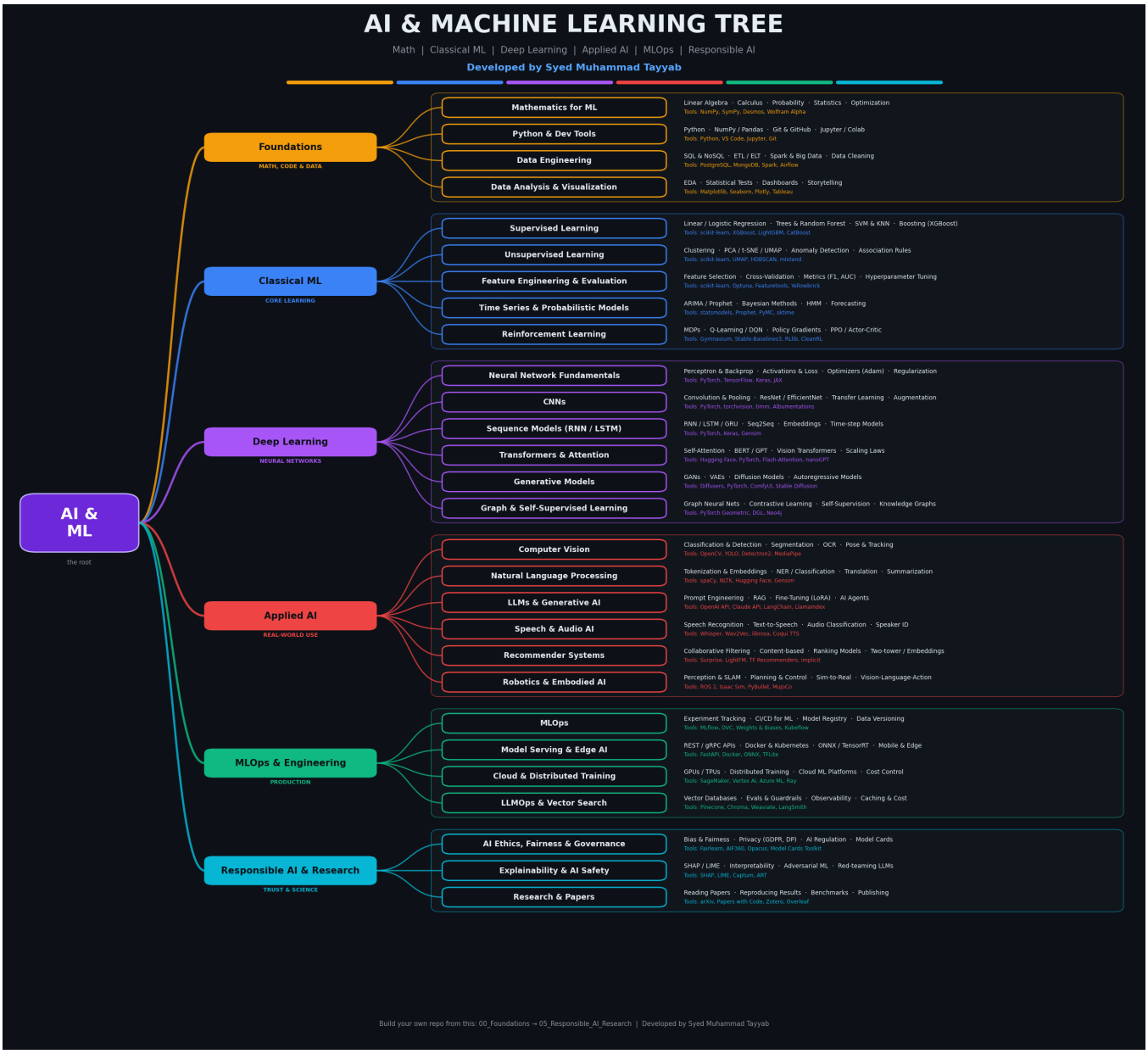
Educational reference. Build, test and deploy AI responsibly.

Developed by Syed Muhammad Tayyab

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1. The Full Tree

Root, 6 branches and 28 fields with their sub-fields and tools.



Branch	Type	What it does	Fields
[ ] **Foundations**	Math, Code & Data	Math, code and data: the base of every model	4
[ ] **Classical ML**	Core Learning	Learn patterns from data without deep nets	5
[ ] **Deep Learning**	Neural Networks	Layers, attention and generative models	6
[ ] **Applied AI**	Real-World Use	Vision, language, speech, recommenders and robots	6
[ ] **MLOps & Engineering**	Production	Ship, scale and monitor models	4
[ ] **Responsible AI & Research**	Trust & Science	Ethics, safety, explainability and papers	3

2. Foundations

Math, Code & Data | Math, code and data: the base of every model

Mathematics for ML `00_Foundations/Mathematics/`

The language models are written in. | Level ★★☆☆☆ | ~6-8 wks

Sub-fields	Linear Algebra | Calculus | Probability | Statistics | Optimization
Tools	NumPy, SymPy, Desmos, Wolfram Alpha
Learn	Khan Academy, MIT 18.06, 3Blue1Brown

Python & Dev Tools `00_Foundations/Python_Dev_Tools/`

Code, notebooks and version control for every experiment. | Level ★☆☆☆☆ | ~3-4 wks

Sub-fields	Python | NumPy / Pandas | Git & GitHub | Jupyter / Colab
Tools	Python, VS Code, Jupyter, Git, Conda
Learn	CS50P, Python for Everybody

Data Engineering `00_Foundations/Data_Engineering/`

Collect, clean, store and move data reliably. | Level ★★★☆☆ | ~5-6 wks

Sub-fields	SQL & NoSQL | ETL / ELT | Spark & Big Data | Data Cleaning
Tools	PostgreSQL, MongoDB, Spark, Airflow, dbt
Learn	Google Data Analytics, Databricks Associate

Data Analysis & Visualization `00_Foundations/Data_Analysis_Viz/`

Understand data before modeling it. | Level ★★☆☆☆ | ~3-4 wks

Sub-fields	EDA | Statistical Tests | Dashboards | Storytelling
Tools	Matplotlib, Seaborn, Plotly, Tableau, Power BI
Learn	IBM Data Analyst, Kaggle Data Viz

3. Classical ML

Core Learning | Learn patterns from data without deep nets

Supervised Learning `01_Classical_ML/Supervised_Learning/`

Learn from labeled examples to predict outcomes. | Level ★★★☆☆ | ~6-8 wks

Sub-fields	Linear / Logistic Regression | Trees & Random Forest | SVM & KNN | Boosting (XGBoost)
Tools	scikit-learn, XGBoost, LightGBM, CatBoost
Learn	Andrew Ng ML Specialization, Kaggle Intro to ML

Unsupervised Learning `01_Classical_ML/Unsupervised_Learning/`

Find structure in data without labels. | Level ★★★☆☆ | ~3-4 wks

Sub-fields	Clustering | PCA / t-SNE / UMAP | Anomaly Detection | Association Rules
Tools	scikit-learn, UMAP, HDBSCAN, mlxtend
Learn	Unsupervised Learning (Coursera)

Feature Engineering & Evaluation `01_Classical_ML/Feature_Eng_Evaluation/`

Turn raw data into signal and measure models honestly. | Level ★★★☆☆ | ~3-4 wks

Sub-fields	Feature Selection | Cross-Validation | Metrics (F1, AUC) | Hyperparameter Tuning
Tools	scikit-learn, Optuna, Featuretools, Yellowbrick
Learn	Kaggle Feature Engineering

Time Series & Probabilistic Models `01_Classical_ML/Time_Series_Probabilistic/`

Forecast the future and reason under uncertainty. | Level ★★★★☆ | ~4-5 wks

Sub-fields	ARIMA / Prophet | Bayesian Methods | HMM | Forecasting
Tools	statsmodels, Prophet, PyMC, sktime
Learn	Kaggle Time Series

Reinforcement Learning `01_Classical_ML/Reinforcement_Learning/`

Agents that learn by trial, reward and policy. | Level ★★★★★ | ~6-8 wks

Sub-fields	MDPs | Q-Learning / DQN | Policy Gradients | PPO / Actor-Critic
Tools	Gymnasium, Stable-Baselines3, RLlib, CleanRL
Learn	David Silver RL, HF Deep RL Course

4. Deep Learning

Neural Networks | Layers, attention and generative models

Neural Network Fundamentals `02_Deep_Learning/Neural_Net_Fundamentals/`

Neurons, layers and how networks learn. | Level ★★★☆☆ | ~5-6 wks

Sub-fields	Perceptron & Backprop | Activations & Loss | Optimizers (Adam) | Regularization
Tools	PyTorch, TensorFlow, Keras, JAX
Learn	Deep Learning Specialization, fast.ai Part 1

CNNs `02_Deep_Learning/CNNs/`

Convolutional networks for grids of pixels. | Level ★★★☆☆ | ~3-4 wks

Sub-fields	Convolution & Pooling | ResNet / EfficientNet | Transfer Learning | Augmentation
Tools	PyTorch, torchvision, timm, Albumentations
Learn	CS231n

Sequence Models (RNN / LSTM) `02_Deep_Learning/Sequence_Models/`

Models that remember what came before. | Level ★★★★☆ | ~2-3 wks

Sub-fields	RNN / LSTM / GRU | Seq2Seq | Embeddings | Time-step Models
Tools	PyTorch, Keras, Gensim
Learn	Sequence Models (Coursera)

Transformers & Attention `02_Deep_Learning/Transformers_Attention/`

The architecture behind modern AI. | Level ★★★★☆ | ~5-6 wks

Sub-fields	Self-Attention | BERT / GPT | Vision Transformers | Scaling Laws
Tools	Hugging Face, PyTorch, Flash-Attention, nanoGPT
Learn	Karpathy: Zero to Hero, CS224N

Generative Models `02_Deep_Learning/Generative_Models/`

Create images, audio, video and text. | Level ★★★★★ | ~5-6 wks

Sub-fields	GANs | VAEs | Diffusion Models | Autoregressive Models
Tools	Diffusers, PyTorch, ComfyUI, Stable Diffusion
Learn	HF Diffusion Course

Graph & Self-Supervised Learning `02_Deep_Learning/Graph_SelfSupervised/`

Learn from relationships and from unlabeled data. | Level ★★★★★ | ~4-5 wks

Sub-fields	Graph Neural Nets | Contrastive Learning | Self-Supervision | Knowledge Graphs
Tools	PyTorch Geometric, DGL, Neo4j
Learn	CS224W

5. Applied AI

Real-World Use | Vision, language, speech, recommenders and robots

Computer Vision `03_Applied_AI/Computer_Vision/`

Teach machines to see. | Level ★★★★☆ | ~6-8 wks

Sub-fields	Classification & Detection | Segmentation | OCR | Pose & Tracking
Tools	OpenCV, YOLO, Detectron2, MediaPipe, SAM
Learn	CS231n, OpenCV University

Natural Language Processing `03_Applied_AI/NLP/`

Teach machines to read and write. | Level ★★★☆☆ | ~6-8 wks

Sub-fields	Tokenization & Embeddings | NER / Classification | Translation | Summarization
Tools	spaCy, NLTK, Hugging Face, Gensim
Learn	CS224N, HF NLP Course

LLMs & Generative AI `03_Applied_AI/LLMs_GenAI/`

Prompting, RAG, fine-tuning and agents. | Level ★★★★☆ | ~6-8 wks

Sub-fields	Prompt Engineering | RAG | Fine-Tuning (LoRA) | AI Agents
Tools	OpenAI API, Claude API, LangChain, LlamaIndex, Ollama
Learn	DeepLearning.AI GenAI, HF LLM Course

Speech & Audio AI `03_Applied_AI/Speech_Audio/`

Hear, transcribe and speak. | Level ★★★★☆ | ~3-4 wks

Sub-fields	Speech Recognition | Text-to-Speech | Audio Classification | Speaker ID
Tools	Whisper, Wav2Vec, librosa, Coqui TTS
Learn	HF Audio Course

Recommender Systems `03_Applied_AI/Recommender_Systems/`

Predict what people will want next. | Level ★★★★☆ | ~4-5 wks

Sub-fields	Collaborative Filtering | Content-based | Ranking Models | Two-tower / Embeddings
Tools	Surprise, LightFM, TF Recommenders, implicit
Learn	Recsys Specialization (UMN)

Robotics & Embodied AI `03_Applied_AI/Robotics_Embodied/`

AI that acts in the physical world. | Level ★★★★★ | ~8-10 wks

Sub-fields	Perception & SLAM | Planning & Control | Sim-to-Real | Vision-Language-Action
Tools	ROS 2, Isaac Sim, PyBullet, MuJoCo
Learn	Modern Robotics (Coursera)

6. MLOps & Engineering

Production | Ship, scale and monitor models

MLOps `04_MLOps_Engineering/MLOps/`

Reliable, repeatable ML from notebook to production. | Level ★★★★☆ | ~5-6 wks

Sub-fields	Experiment Tracking | CI/CD for ML | Model Registry | Data Versioning
Tools	MLflow, DVC, Weights & Biases, Kubeflow, GitHub Actions
Learn	MLOps Zoomcamp, Google Prof. ML Engineer

Model Serving & Edge AI `04_MLOps_Engineering/Serving_Edge/`

Put models behind APIs and onto devices. | Level ★★★☆☆ | ~3-4 wks

Sub-fields	REST / gRPC APIs | Docker & Kubernetes | ONNX / TensorRT | Mobile & Edge
Tools	FastAPI, Docker, ONNX, TFLite, BentoML
Learn	AWS ML Specialty

Cloud & Distributed Training `04_MLOps_Engineering/Cloud_Distributed/`

Train big models on big hardware. | Level ★★★★☆ | ~4-5 wks

Sub-fields	GPUs / TPUs | Distributed Training | Cloud ML Platforms | Cost Control
Tools	SageMaker, Vertex AI, Azure ML, Ray, DeepSpeed
Learn	Azure DP-100, AWS ML Specialty

LLMOps & Vector Search `04_MLOps_Engineering/LLMOps_Vector_Search/`

Run LLM apps reliably and cheaply. | Level ★★★★☆ | ~3-4 wks

Sub-fields	Vector Databases | Evals & Guardrails | Observability | Caching & Cost
Tools	Pinecone, Chroma, Weaviate, LangSmith, Ragas
Learn	DeepLearning.AI LLMOps

7. Responsible AI & Research

Trust & Science | Ethics, safety, explainability and papers

AI Ethics, Fairness & Governance `05_Responsible_AI_Research/Ethics_Fairness_Governance/`

Build AI people can trust. | Level ★★☆☆☆ | ~2-3 wks

Sub-fields	Bias & Fairness | Privacy (GDPR, DP) | AI Regulation | Model Cards
Tools	Fairlearn, AIF360, Opacus, Model Cards Toolkit
Learn	Elements of AI, AI Ethics (Coursera)

Explainability & AI Safety `05_Responsible_AI_Research/Explainability_Safety/`

Understand, test and harden models. | Level ★★★★☆ | ~3-4 wks

Sub-fields	SHAP / LIME | Interpretability | Adversarial ML | Red-teaming LLMs
Tools	SHAP, LIME, Captum, ART, Garak
Learn	AI Safety Fundamentals

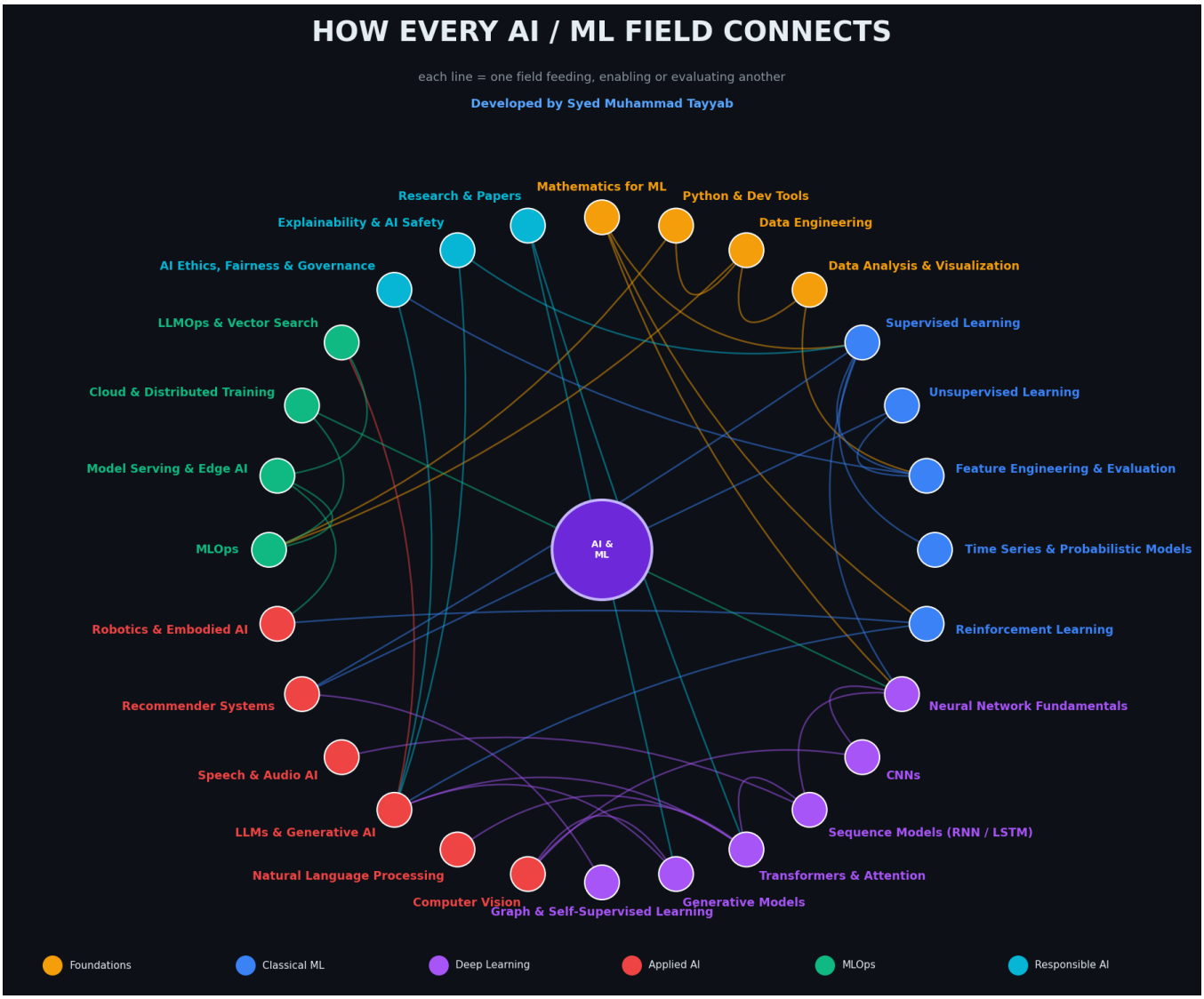
Research & Papers `05_Responsible_AI_Research/Research_Papers/`

Read, reproduce and publish. | Level ★★★☆☆ | ~ongoing wks

Sub-fields	Reading Papers | Reproducing Results | Benchmarks | Publishing
Tools	arXiv, Papers with Code, Zotero, Overleaf, W&B
Learn	Research Methods

8. How Everything Is Linked

Each line = one field feeding, protecting or testing another.



8.1 Link Table

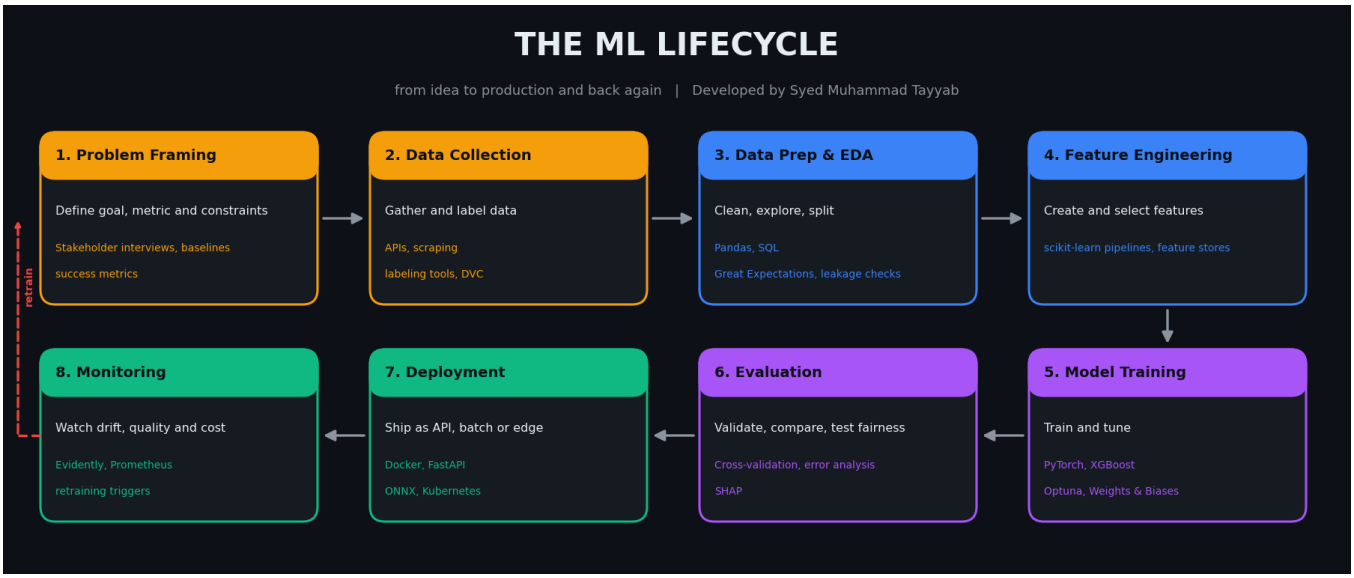
From		To	Why they connect
Mathematics for ML	->	Supervised Learning	stats & optimization power every model
Mathematics for ML	->	Neural Network Fundamentals	calculus & linear algebra drive backprop
Mathematics for ML	->	Reinforcement Learning	probability & Markov decision processes
Python & Dev Tools	->	Data Engineering	pipelines written in Python & SQL
Python & Dev Tools	->	MLOps	automation & testing
Data Engineering	->	Data Analysis & Visualization	clean data enables insight
Data Analysis & Visualization	->	Feature Engineering & Evaluation	insights guide features
Data Engineering	->	MLOps	data versioning & pipelines
Feature Engineering & Evaluation	->	Supervised Learning	better features, better models
Unsupervised Learning	->	Feature Engineering & Evaluation	embeddings & dimensionality reduction
Time Series & Probabilistic Models	->	Supervised Learning	forecasting as supervised learning
Supervised Learning	->	Neural Network Fundamentals	neural nets generalize classic models
Neural Network Fundamentals	->	CNNs	convolutions for images
Neural Network Fundamentals	->	Sequence Models (RNN / LSTM)	recurrence for sequences
Sequence Models (RNN / LSTM)	->	Transformers & Attention	attention replaces recurrence
Transformers & Attention	->	LLMs & Generative AI	foundation of LLMs
Transformers & Attention	->	Natural Language Processing	BERT-style language models
CNNs	->	Computer Vision	vision backbones
Transformers & Attention	->	Computer Vision	vision transformers
Generative Models	->	Computer Vision	image generation & editing
Generative Models	->	LLMs & Generative AI	autoregressive text generation
Reinforcement Learning	->	Robotics & Embodied AI	policies for control
Reinforcement Learning	->	LLMs & Generative AI	RLHF aligns assistants
Sequence Models (RNN / LSTM)	->	Speech & Audio AI	speech recognition models
Supervised Learning	->	Recommender Systems	ranking & prediction models
Graph & Self-Supervised Learning	->	Recommender Systems	graph-based recommendations
Unsupervised Learning	->	Recommender Systems	embeddings & user clustering
LLMs & Generative AI	->	LLMOps & Vector Search	RAG, evals & guardrails
LLMOps & Vector Search	->	Model Serving & Edge AI	low-latency LLM serving
MLOps	->	Model Serving & Edge AI	CI/CD into deployment
Cloud & Distributed Training	->	Neural Network Fundamentals	GPU & distributed training
Cloud & Distributed Training	->	MLOps	scalable infrastructure
Model Serving & Edge AI	->	Robotics & Embodied AI	edge & on-device inference

From		To	Why they connect
Feature Engineering & Evaluation	->	AI Ethics, Fairness & Governance	metrics reveal bias
Explainability & AI Safety	->	Supervised Learning	explain predictions (SHAP / LIME)
Explainability & AI Safety	->	LLMs & Generative AI	jailbreaks & prompt injection
AI Ethics, Fairness & Governance	->	LLMs & Generative AI	safety & alignment
Research & Papers	->	Transformers & Attention	papers drive new architectures
Research & Papers	->	Generative Models	new generative methods

9. The ML Lifecycle

From idea to production and back again: where every field plugs in.

Stage	What happens	Tools & good practice
1. Problem Framing	Define goal, metric and constraints	Stakeholder interviews, baselines, success metrics
2. Data Collection	Gather and label data	APIs, scraping, labeling tools, DVC
3. Data Prep & EDA	Clean, explore, split	Pandas, SQL, Great Expectations, leakage checks
4. Feature Engineering	Create and select features	scikit-learn pipelines, feature stores
5. Model Training	Train and tune	PyTorch, XGBoost, Optuna, Weights & Biases
6. Evaluation	Validate, compare, test fairness	Cross-validation, error analysis, SHAP
7. Deployment	Ship as API, batch or edge	Docker, FastAPI, ONNX, Kubernetes
8. Monitoring	Watch drift, quality and cost	Evidently, Prometheus, retraining triggers



Types of AI Work

Foundations (Amber)	Math, Python and data: the base of everything.
Classical ML (Blue)	Predict, cluster and evaluate with classic algorithms.
Deep Learning (Purple)	Neural nets, attention and generative models.
Applied AI (Red)	Vision, language, speech, recommenders and robots.
MLOps (Green)	Ship, scale, serve and monitor models.
Responsible AI (Cyan)	Ethics, safety, explainability and research.

10. Folder Structure (your study repo)

Which field lives in which folder. Each leaf folder holds notes/, labs/, tools/ and cheatsheets/.

```
AI-ML-Guide/
├── 00_Foundations/   [Math, Code & Data]
│   ├── Mathematics/
│   ├── Python_Dev_Tools/
│   ├── Data_Engineering/
│   └── Data_Analysis_Viz/
├── 01_Classical_ML/   [Core Learning]
│   ├── Supervised_Learning/
│   ├── Unsupervised_Learning/
│   ├── Feature_Eng_Evaluation/
│   ├── Time_Series_Probabilistic/
│   └── Reinforcement_Learning/
├── 02_Deep_Learning/   [Neural Networks]
│   ├── Neural_Net_Fundamentals/
│   ├── CNNs/
│   ├── Sequence_Models/
│   ├── Transformers_Attention/
│   ├── Generative_Models/
│   └── Graph_SelfSupervised/
├── 03_Applied_AI/   [Real-World Use]
│   ├── Computer_Vision/
│   ├── NLP/
│   ├── LLMs_GenAI/
│   ├── Speech_Audio/
│   ├── Recommender_Systems/
│   └── Robotics_Embodied/
├── 04_MLOps_Engineering/   [Production]
│   ├── MLOps/
│   ├── Serving_Edge/
│   ├── Cloud_Distributed/
│   └── LLMOps_Vector_Search/
├── 05_Responsible_AI_Research/   [Trust & Science]
│   ├── Ethics_Fairness_Governance/
│   ├── Explainability_Safety/
│   └── Research_Papers/
├── 06_Projects/   # Portfolio: end-to-end ML apps
├── 07_Kaggle_Competitions/   # Notebooks, solutions, leaderboard notes
├── 08_Papers_Notes/   # Paper summaries and reproductions
└── 09_Resources/   # Books, courses, datasets, cheat sheets
```

11. Learning & Certification Roadmap

Pick a track after the foundations. Most people start with Andrew Ng's ML Specialization or Kaggle Learn.

Track	Certifications
Beginner	Andrew Ng ML Specialization, Kaggle Learn, fast.ai, CS50 AI
Data Scientist	IBM Data Science, Google Data Analytics, Kaggle Competitions, Azure DP-100
ML Engineer	AWS ML Specialty, Google Prof. ML Engineer, MLOps Zoomcamp
Research / GenAI	Deep Learning Specialization, Hugging Face Course, CS224N, CS231n

Suggested order

1	**Foundations**	Math, Python, SQL and data analysis
2	**Classical ML**	Models, features, metrics, Kaggle
3	**Deep Learning**	PyTorch, CNNs, Transformers
4	**Specialize**	NLP, vision, LLMs, RL or recommenders
5	**Ship & Share**	MLOps, deployed projects, papers, GitHub

Portfolio Project Ideas

Level	Project	Skills
Beginner	House price predictor / Titanic on Kaggle	EDA, regression, scikit-learn
Beginner	Spam or sentiment classifier	NLP basics, evaluation
Intermediate	Image classifier with transfer learning	CNNs, PyTorch, augmentation
Intermediate	Movie recommender with a web UI	Recsys, FastAPI, React
Advanced	RAG chatbot over your own documents	LLMs, vector DB, evals
Advanced	End-to-end pipeline with CI/CD and drift monitoring	MLOps, Docker, MLflow